

CA-AFTER-MID — Weeks 9–15 · 12 Oct – 29 Nov 2026 · FINAL (W15)

Scope: Weeks 9–15 (12 Oct – 29 Nov 2026) — ILP & compiler techniques, VLIW/superscalar, SIMD/GPU, memory hierarchy & cache design, virtual memory & TLB, RAID & advanced topics, ending at the Week 15 final. **Siblings:** `CA-NAV.md` (master map) · `CA-BEFORE-MID.md` (Weeks 1–8). Week-colored headers below match the NAV schedule. **Exam split:** 60% diagram accuracy + numerical precision, 40% theory. Drawing from memory is the skill. **Exam order:** numericals → diagrams → theory (final).

At a glance (W9–W15)

W	Dates	Variant	File	Topics	Chapter
W9	12–18 Oct	P1	AFTER	ILP & compiler techniques (recovery)	— (pack-level)
W10	19–25 Oct	P1	AFTER	VLIW, superscalar, SIMD/GPU	— (pack-level)
W11	26 Oct–01 Nov	P0	AFTER	Memory hierarchy & cache design	05
W12	02–08 Nov	P0	AFTER	Virtual memory & TLB	05
W13	09–15 Nov	P1	AFTER	RAID & advanced topics	— (pack-level)
W14	16–22 Nov	P0	AFTER	Final taper — past paper FIRST	01–05
W15	23–29 Nov	FINAL	AFTER	Exam — execute	—

Tier key: **P0** = no exam pressure (front-load new material) · **P1** = light revision · **P2** = drill-heavy week.

W9 — ILP & Compiler Techniques · 12–18 Oct · P1 · 6 hrs

Banner: Rotation CA (Mon+Tue) · Tier P1 · Time budget 6 hrs · P0 floor: pack pass + def/formula skim + same-problem drill.
Recovery week — ledger MUST clear to 0.

Sources & offsets

Source	Where	Ref
Narrative	<code>Week-by-Week-Narrative.md</code> → Week 9	lines 152–171
Fear-Killer Pack	<code>Fear-Killer-Packs.md</code> → CA-W9	lines 71–77
Week manifest	<code>weeks/CA-W9.md</code>	full file
Chapter breakdown	— (M5 gap: compiler ILP + branch prediction is pack-level only)	—

Definitions (verbatim)

Term	Definition
Amdahl's Law	The speedup of a system using multiple processors is limited by the fraction of the task that cannot be parallelized. $\text{Speedup} = 1 / ((1 - f) + f/n)$.
ILP (Instruction-Level Parallelism)	The degree to which instructions in a program can be executed in parallel.

Formulas (verbatim)

Formula	Statement
Amdahl's Law	$\text{Speedup} = 1 / ((1 - f) + f/n)$
CPU Time	$\text{CPU Time} = \text{Instruction Count} \times \text{CPI} \times \text{Cycle Time}$
Speedup	$\text{Speedup} = \text{Old Time} / \text{New Time}$

Type	Items
Diagrams	— (pack-level; reuse pipeline #3/#4 for ILP context)
Numericals	Numerical-Book.md #4 Amdahl's Law · #2 CPU time
Tricky	Top-10-Tricky-Concepts.md #9 Amdahl counterintuitive result
Top-100	Top-100-Questions.md #78 loop unrolling ILP · #79/#84/#88 ILP walls · #82/#83 Amdahl · #90 software pipelining

Books · Chapters · Media

Resource	Where
Hennessy & Patterson CAQA	Ch 3 (ILP: unrolling, scheduling, software pipelining)
Sarangi (IITD)	Ch 10 (forwarding & performance)

Fear-Killer Pack CA-W9 (verbatim)

Resources: H&P CAQA Ch 3 (ILP: unrolling, scheduling, software pipelining)

1. Unroll the following loop 4 times and schedule the resulting code to minimize stalls: `for(i=0; i<N; i++) A[i] = B[i] + C[i]`. Show the original pipeline schedule, the unrolled code, and the final schedule. Calculate the speedup from unrolling.
2. Given `for(i=0; i<100; i++) A[i] = A[i-1] + A[i+1]`, identify all loop-carried dependencies. Can this loop be parallelized across multiple processors? If not, show a transformation that enables parallelization.
3. Convert a 3-cycle loop body into a software-pipelined version showing the prologue, kernel, and epilogue. Take `lw $t0, 0($s1) / add $t0, $t0, $s2 / sw $t0, 0($s1) / addi $s1, $s1, 4` and software-pipeline it.
4. A program is 80% parallelizable and runs on 4 processors. Compute the Amdahl's Law speedup. Now show that even with infinite processors, 10% sequential code caps speedup at 10x. Why do students forget the sequential fraction?

Narrative — Week 9 (verbatim)

Topics: Instruction-Level Parallelism; ILP limitations (Amdahl's Law, ILP wall, power wall, memory wall); Loop unrolling; Software pipelining; Dependence analysis

Resources: Hennessy & Patterson Ch.3; Sarangi Ch. 10 (forwarding & performance)

Recovery note: If midterm scored below 85%, take 2 days at reduced load (20 hrs/week total). Recalibrate — a 78% midterm with 95%+ finals still yields B+/A-.

Practice: - Unroll a loop with 4 iterations and identify remaining dependences — closed book, retrieve from memory - Distinguish loop-carried vs loop-independent dependences — retrieve definitions from blank page - Software pipelining: transform a loop for maximum throughput — closed book - **Killer trap:** Amdahl's Law — $\text{Speedup} = 1 / ((1 - f) + f/n)$. Students forget that speedup is limited by the non-parallelizable fraction f. Even infinite processors cannot speed up the sequential portion. - **Interleaved retrieval (20 min):** Draw the multi-cycle datapath from blank paper in 12 min. Trace R-type, lw, sw, branch through it. Check accuracy only after finishing.

Retrieval protocol: Every practice item starts from a blank page. No peeking at notes. Answers checked only after completion.

Deliverable: 5 ILP numerical problems (Amdahl's Law + speedup calculations) + loop unrolling trace

Time budget: 6 hrs

P0 floor · Drill target · Deliverable · Trap

- **P0 floor:** fear-killer pack pass (Q4 Amdahl first + Q1 unroll) · def/formula skim (ILP def + Amdahl formula) · same-problem drill #4
- **Same-problem drill target:** Amdahl speedup < 30 s; unroll+schedule a loop without stalling
- **Deliverable:** 5 ILP numerical problems (Amdahl's Law + speedup calculations) + loop unrolling trace
- **Trap:** Amdahl's Law — speedup is limited by the non-parallelizable fraction f; even infinite processors can't speed up the sequential portion. Recovery mandate: ledger MUST clear to 0.

Sources & offsets

Source	Where	Ref
Narrative	Week-by-Week-Narrative.md → Week 10	lines 175–189
Fear-Killer Pack	Fear-Killer-Packs.md → CA-W10	lines 79–86
Week manifest	weeks/CA-W10.md	full file
Chapter breakdown	— (M6/M7 gap: VLIW/superscalar/ILP limits; SIMD/GPU is pack-level only)	—

Definitions (verbatim)

Term	Definition
Superscalar	A processor that can issue multiple instructions per clock cycle.
VLIW (Very Long Instruction Word)	An architecture where multiple operations are encoded in a single long instruction.

Formulas (verbatim)

Formula	Statement
CPU Time	CPU Time = Instruction Count × CPI × Cycle Time
Speedup	Speedup = Old Time / New Time
Effective IPC	Instructions completed per cycle — capped below issue width by stalls + branch mispredictions.

Diagrams · Numericals · Tricky · Top-100

Type	Items
Diagrams	Diagram-Book.md #14 VLIW instruction packet · #15 superscalar pipeline (canonical)
Numericals	Numerical-Book.md #2 CPU time
Tricky	Top-10-Tricky-Concepts.md #10 VLIW vs Superscalar trade-off
Top-100	Top-100-Questions.md #76 VLIW vs Superscalar · #77 SIMD vs SIMT · #85 GPU warp scheduling · #86 memory coalescing · #87 VLIW static scheduling

Books · Chapters · Media

Resource	Where
Hennessy & Patterson CAQA	Ch 3.6–3.7 (superscalar/VLIW) · Ch 4 (SIMD/GPU)
Patterson & Hennessy COD	Ch 6 + App B (GPUs)
Sarangi (IITD)	Ch 12
Media	UC Berkeley CS61C · NVIDIA CUDA basics

Fear-Killer Pack CA-W10 (verbatim)

Resources: H&P CAQA Ch 3.6-3.7 (superscalar/VLIW), Ch 4 (SIMD/GPU) • P&H COD Ch 6 + App B (GPUs) • UC Berkeley CS61C

- Design a VLIW instruction packet for a 2-issue processor (one ALU, one memory/ALU). Schedule `lw $t0,0($t1) / add $t2,$t3,$t4 / sub $t5,$t6,$t7 / sw $t5,0($t8)` into VLIW packets. How many cycles vs a 2-issue in-order superscalar?
- A superscalar issues 4 instructions per cycle with a 5-cycle branch misprediction penalty. For a program with 20% branches and 90% prediction accuracy, compute effective IPC. Explain why IPC can never reach 4.
- Write a SIMD addition of two 8-element arrays using a hypothetical 256-bit SIMD unit (8×32-bit). Then write the equivalent scalar loop. For N=1024, calculate the instruction count saved. Now show what happens when elements are accessed with a stride of 2 — can you still vectorize?
- Explain the SIMT execution model used in NVIDIA GPUs. A warp of 32 threads encounters an if/else branch where 16 threads take the `if` path and 16 take `else`. Show the execution timeline and compute utilization. How does warp divergence affect performance?
- Draw the GPU memory hierarchy showing global, shared, local, constant, and register memory with latency and scope of

each. Given a matrix multiplication kernel, decide which memory type to use for: tile data, row pointers, constants, per-thread accumulators.

Narrative — Week 10 (verbatim)

Topics: VLIW architecture; Superscalar processors; In-order vs out-of-order; SIMD vs SIMT; GPU architecture

Resources: Sarangi Ch. 12; NVIDIA CUDA basics

Practice: - VLIW vs Superscalar comparison: static vs dynamic scheduling — closed book - GPU architecture: warps, thread blocks, memory coalescing — retrieve from memory - SIMT vs SIMD: when does each excel? - **Interleaved retrieval (20 min):** 5 pipeline hazard sequences from blank paper — identify RAW/WAW/WAR, draw forwarding paths, show stall vs forward decisions. All closed book.

Deliverable: VLIW vs Superscalar comparison table + GPU architecture summary

Time budget: 6 hrs

P0 floor · Drill target · Deliverable · Trap

- **P0 floor:** fear-killer pack pass (Q1 VLIW packet + Q4 SIMT timeline) · def/formula skim (Superscalar/VLIW defs + Performance formulas) · same-problem drill #2
- **Same-problem drill target:** VLIW packet scheduling < 30 s; warp-divergence utilization calc < 2 min
- **Deliverable:** VLIW vs Superscalar comparison table + GPU architecture summary
- **Trap:** M6/M7 gap — no chapter breakdown for VLIW/superscalar or SIMD/GPU; mapped at pack level

W11 — Memory Hierarchy & Cache Design · 26 Oct–01 Nov · P0 · 6 hrs

Banner: Rotation CA (Mon+Tue) · Tier P0 · Time budget 6 hrs · P0 floor: pack pass + def/formula skim + same-problem drill.

Sources & offsets

Source	Where	Ref
Narrative	Week-by-Week-Narrative.md → Week 11	lines 193–207
Fear-Killer Pack	Fear-Killer-Packs.md → CA-W11	lines 88–93
Week manifest	weeks/CA-W11.md	full file
Chapter breakdown	03-Chapter-Breakdowns/05-... (cache part)	embedded below

Definitions (verbatim)

Term	Definition
Cache	A small, fast memory that stores frequently accessed data to reduce average memory access time.
Direct-Mapped Cache	A cache organization where each memory block maps to exactly one cache line.
Set-Associative Cache	A cache where each memory block maps to a set of multiple cache lines.
AMAT (Average Memory Access Time)	The average time to access memory, computed as Hit Time + Miss Rate × Miss Penalty.

Formulas (verbatim)

Formula	Statement
AMAT	AMAT = Hit Time + Miss Rate × Miss Penalty
Offset bits	Offset bits = log ₂ (Block Size)
Index bits	Index bits = log ₂ (Number of Sets)
Tag bits	Tag bits = Address bits – Index – Offset
Cache Size	Cache Size = Number of Sets × Associativity × Block Size

Type	Items
Diagrams	Diagram-Book.md #10 direct-mapped cache · #11 set-associative cache (canonical)
Numericals	Numerical-Book.md #5 address breakdown · #6 AMAT · #8 miss rate
Tricky	Top-10-Tricky-Concepts.md #7 write-back vs write-through coherence
Top-100	Top-100-Questions.md #56–#75 (cache & memory)

Books · Chapters · Media

Resource	Where
Sarangi (IITD)	Ch 11 (cache) · CA textbook Ch 8
Hennessy & Patterson CAQA	Ch 2 + App B
Patterson & Hennessy COD	Ch 5 (memory hierarchy)

Fear-Killer Pack CA-W11 (verbatim)

Resources: H&P CAQA Ch 2 + App B • P&H COD Ch 5

- A processor has a 32-bit address bus, 16 KB direct-mapped cache with 64-byte blocks. Calculate tag/index/offset sizes. Then trace accesses: 0x0000, 0x0040, 0x0100, 0x0040, 0x0000. Show hit/miss with LRU. Now redesign as 2-way set-associative — how do the bit fields change?
- Calculate AMAT for: L1 hit time = 2 cycles, L1 miss rate = 5%, L2 hit time = 10 cycles, L2 miss rate = 20% (of L1 misses), memory = 200 cycles. Which component contributes the most to access time? What happens if you double L1 size (miss rate drops to 3% but hit time becomes 3 cycles)?
- Compare write-back vs write-through: traffic, coherence, and performance trade-offs. From a given access sequence, compute the miss rate for direct-mapped vs set-associative organizations.

Narrative — Week 11 (verbatim)

Topics: Cache memory; Direct-mapped, set-associative, fully-associative; Write policies; AMAT calculation

Resources: Sarangi Ch. 11; CA Ch.8

Practice: - 10 cache numerical problems: address breakdown, tag/index/offset, miss rate, AMAT — closed book - Direct-mapped vs set-associative: compare miss rates — retrieve from memory - Write-back vs write-through: performance tradeoffs — blank page comparison table - **Interleaved retrieval (20 min):** FSM control unit design from blank page — draw complete state diagram for R-type, lw, sw, branch. Every state transition verified. All closed book.

Deliverable: 5 cache performance calculations

Time budget: 6 hrs

Chapter 5 — Cache and Virtual Memory (verbatim)

Weight: ★★★★★☆

Difficulty: ★★★★★☆

Importance: ★★★★★☆

Learning Objectives

Explain

- Temporal and spatial locality
- Cache mapping techniques
- Virtual memory and paging

Compare

- Direct-mapped vs set-associative vs fully-associative
- Write-back vs write-through

- Tag, index, offset from address
- AMAT (Average Memory Access Time)
- Effective access time with TLB
- Page table size

Design

- Cache organization for given parameters

Must Memorize

AMAT: Average Memory Access Time = Hit Time + Miss Rate $\times$ Miss Penalty

Effective Access Time (with TLB): $EAT = TLB \text{ Hit Rate} \times (TLB \text{ Hit Time} + \text{Cache Hit Time}) + (1 - TLB \text{ Hit Rate}) \times (TLB \text{ Miss Penalty} + \text{Cache Access Time})$

Must Practice

- 10+ cache address breakdown problems
- 10+ AMAT calculations
- 5+ page table/TLB problems

Common Mistakes

1. **Address breakdown** — bits for offset = $\log_2(\text{block size})$, index = $\log_2(\text{number of sets})$, tag = remaining
2. **Write-back vs write-through distinction** — write-back updates cache only, writes to memory only on eviction
3. **TLB reach** = TLB entries $\times$ page size
4. **Multi-level page tables** — each level adds an extra memory access

GPA Priority: ■ High Yield

P0 floor · Drill target · Deliverable · Trap

- **P0 floor:** fear-killer pack pass (Q1 tag/index/offset + Q2 AMAT) · def/formula skim (Cache/Direct-Mapped/Set-Associative defs + Cache formulas) · same-problem drill #5/#6/#8
- **Same-problem drill target:** AMAT < 30 s; address breakdown (tag/index/offset) < 1 min
- **Deliverable:** 5 cache performance calculations
- **Trap:** offset = $\log_2(\text{block size})$, index = $\log_2(\text{number of sets})$, tag = remaining bits

W12 — Virtual Memory & TLB · 02–08 Nov · P0 · 6 hrs

Banner: Rotation CA (Mon+Tue) · Tier P0 · Time budget 6 hrs · P0 floor: pack pass + def/formula skim + same-problem drill.
Start sleep banking: 9 hrs (bedtime 22:00) for 7 nights.

Sources & offsets

Source	Where	Ref
Narrative	Week-by-Week-Narrative.md → Week 12	lines 211–228
Fear-Killer Pack	Fear-Killer-Packs.md → CA-W12	lines 95–100
Week manifest	weeks/CA-W12.md	full file
Chapter breakdown	03-Chapter-Breakdowns/05-... (VM part)	reference (embedded W11)

Definitions (verbatim)

Term	Definition
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Virtual Memory	A memory management technique that gives each process the illusion of a large, contiguous address space.
TLB (Translation Lookaside Buffer)	A cache for page table entries that speeds up virtual-to-physical address translation.

Formulas (verbatim)

Formula	Statement
EAT (with TLB)	$EAT = H \times (t + m) + (1 - H) \times (t + k \cdot n + m)$ — H = TLB hit rate, t = TLB access time, k = page-table levels, n = memory access per level, m = data memory access time
EAT (simplified)	$EAT = H \times m + (1 - H) \times 2m$
TLB Reach	$TLB\ Reach = TLB\ Entries \times Page\ Size$
Page Table Size	$Page\ Table\ Size = Number\ of\ Pages \times Page\ Table\ Entry\ Size$

Diagrams · Numericals · Tricky · Top-100

Type	Items
Diagrams	Diagram-Book.md #12 virtual → physical translation (canonical) · #17 two-level page table walk · #19 TLB structure
Numericals	Numerical-Book.md #9 EAT · #10 TLB reach · #11 page table size · #12 two-level paging
Tricky	— (pack-level; see Chapter 5 common mistakes #3/#4)
Top-100	Top-100-Questions.md #66–#70 · #75 (VM/TLB)

Books · Chapters · Media

Resource	Where
Patterson & Hennessy COD	Ch 5 (VM/TLB)
Hennessy & Patterson CAQA	Ch 2 (+ online App L)
Sarangi (IITD)	Ch 11 (Virtual Memory)
OS concepts	overlap with CS-329

Fear-Killer Pack CA-W12 (verbatim)

Resources: P&H COD Ch 5 (VM/TLB) • H&P CAQA Ch 2 (+ online App L) • OS concepts overlap with CS-329

- Given TLB access time = 1 cycle, memory access = 100 cycles, TLB hit rate = 98%, page fault rate = 1% of TLB misses, fault time = 5ms, calculate effective access time. Which component dominates? What if TLB hit rate drops to 90%?
- For a 32-bit address with 4 KB pages, 10-bit first-level page table index, and 10-bit second-level index, compute the offset bits. Calculate total page table size for a process using 4 MB of memory. How does this change with 2 MB pages?
- Compute TLB reach for 64 entries with 4 KB pages, then with 2 MB pages. Walk a two-level page table for a given virtual address showing the address split and each memory access.

Narrative — Week 12 (verbatim)

Topics: Virtual memory; Page tables; TLB; Effective access time; Multi-level paging

Resources: Sarangi Ch. 11 (Virtual Memory); OS concepts overlap with CS-329

Sleep banking: Begin sleeping 9 hrs (bedtime 22:00) for 7 nights leading to finals. This inoculates against exam-night sleep loss.

Practice: - Page table size calculation for various architectures — closed book - TLB reach computation — retrieve formula from memory - Effective access time formula with TLB hit rate — derive from first principles - Two-level paging: walk through a virtual address translation — from blank page - **Interleaved retrieval (20 min):** Microprogrammed control — compare horizontal vs vertical from memory. Then single-cycle datapath from blank paper in 10 min. These topics (W1, W4) have had zero retrieval since midterm — retrieve before they decay below threshold.

Deliverable: 5 virtual memory numerical problems

Time budget: 6 hrs

P0 floor · Drill target · Deliverable · Trap

- **P0 floor:** fear-killer pack pass (Q1 EAT + Q3 TLB reach) · def/formula skim (TLB/Virtual Memory defs + VM formulas) · same-problem drill #9/#10/#11/#12
- **Same-problem drill target:** EAT < 30 s; page-table size calc < 1 min; TLB reach < 30 s
- **Deliverable:** 5 virtual memory numerical problems
- **Trap:** multi-level page tables — each level adds an extra memory access; W1/W4 topics have had zero retrieval since midterm — retrieve before decay

W13 — RAID & Advanced Topics · 09–15 Nov · P1 · 5 hrs

Banner: Rotation CA (Mon+Tue) · Tier P1 · Time budget 5 hrs · P0 floor: pack pass + def/formula skim + same-problem drill.

Sources & offsets

Source	Where	Ref
Narrative	Week-by-Week-Narrative.md → Week 13	lines 232–246
Fear-Killer Pack	Fear-Killer-Packs.md → CA-W13	lines 102–107
Week manifest	weeks/CA-W13.md	full file
Chapter breakdown	— (M9 RAID is pack-only; Ch5 covers cache+VM)	reference (embedded W11)

Definitions (verbatim)

Term	Definition
RAID	Redundant Array of Independent Disks — a storage technology that combines multiple disk drives for performance and/or redundancy.

Formulas (verbatim)

Formula	Statement
RAID 5 Usable	RAID 5 Usable = (N – 1) × Disk Size
RAID 5 Write Penalty	RAID 5 Write Penalty = 4 I/Os per write (2 reads + 2 writes)

Diagrams · Numericals · Tricky · Top-100

Type	Items
Diagrams	Diagram-Book.md #13 RAID 0, 1, 5, 6, 10 layouts (canonical)
Numericals	Numerical-Book.md #6 AMAT (for cache inclusion) · #24 write buffer depth · #25 multi-level cache AMAT
Tricky	Top-10-Tricky-Concepts.md #8 RAID 4 vs 5 write penalty
Top-100	Top-100-Questions.md #80 RAID 5 write penalty · #81 RAID levels · #89 RAID capacity

Books · Chapters · Media

Resource	Where
Patterson & Hennessey COD	Ch 5.11 (RAID)
Hennessey & Patterson CAQA	online App D (storage systems)
Sarangi (IITD)	Ch 13 (I/O & Storage — RAID)

Fear-Killer Pack CA-W13 (verbatim)

Resources: P&H COD Ch 5.11 (RAID) • H&P CAQA online App D (storage systems)

1. A RAID 5 array has 8 disks (1 parity, 7 data). Compute the write penalty for a 4 KB write. Now compare RAID 5 vs RAID 10 for a write-heavy workload (70% writes, random 4 KB). Calculate IOPS for each.
2. Build the RAID comparison table from memory: RAID 0, 1, 4, 5, 6, 10 — capacity, redundancy, read/write performance. Why is RAID 4's dedicated parity disk a bottleneck that RAID 5 fixes?

3. Explain multi-level cache inclusion policies and I/O system performance. How does a write buffer depth affect throughput given a write frequency?

Narrative — Week 13 (verbatim)

Topics: RAID levels 0, 1, 4, 5, 6, 10; RAID performance; Advanced memory design; I/O systems

Resources: Sarangi Ch. 13 (I/O & Storage — RAID); textbook RAID section

Practice: - RAID comparison table: capacity, redundancy, read/write performance — retrieve from memory - RAID 4 vs 5 write penalty calculation — closed book - Multi-level cache inclusion policies — blank page comparison - **Interleaved retrieval (20 min)**: Timed multi-cycle datapath draw in 10 min from blank paper. Then 3 pipeline hazard sequence traces. This is your final speed drill before finals.

Deliverable: RAID comparison table + multi-level cache inclusion policy analysis

Time budget: 5 hrs

P0 floor · Drill target · Deliverable · Trap

- **P0 floor**: fear-killer pack pass (Q1 write penalty + Q2 RAID table) · def/formula skim (RAID def + RAID formulas) · same-problem drill #6
- **Same-problem drill target**: RAID write penalty < 30 s; RAID table from memory in full
- **Deliverable**: RAID comparison table + multi-level cache inclusion policy analysis
- **Trap**: RAID 4's dedicated parity disk is a bottleneck — RAID 5 distributes parity (write penalty = 4 I/Os)

W14 — Final Taper · 16–22 Nov · P0 · 8 hrs

Banner: Rotation CA (Mon+Tue) · Tier P0 · Time budget 8 hrs · P0 floor: pack pass + def/formula skim + same-problem drill.
START with the past paper, NOT with review.

Sources & offsets

Source	Where	Ref
Narrative	Week-by-Week-Narrative.md → Week 14	lines 250–269
Fear-Killer Pack	Fear-Killer-Packs.md → CA-W14	lines 109–116
Week manifest	weeks/CA-W14.md	full file
Chapter breakdowns	all 5 files	reference (embedded W1, W2, W5, W11)

Definitions (verbatim) — full W1–W13 sweep

Term	Definition
ISA	The interface between hardware and software, defining the instructions, registers, memory addressing, and data types.
Single-cycle Datapath	A processor where every instruction executes in one clock cycle.
Multi-cycle Datapath	A processor implementation that breaks instruction execution into multiple steps.
Microprogrammed Control	A control unit implementation that uses microinstructions stored in a control store to generate control signals.
Hazard	A condition that prevents the next instruction in the pipeline from executing during its designated clock cycle.
RAW Hazard	A true data dependence where an instruction reads a register that a previous instruction writes (Read After Write).
Forwarding (Bypassing)	A technique that resolves data hazards by feeding the ALU output from a later pipeline stage back to the ALU input of an earlier stage.
Branch Prediction	A technique that predicts the outcome of a conditional branch before it is resolved, allowing the pipeline to continue without stalling.
ILP	The degree to which instructions in a program can be executed in parallel.

Amdahl's Law	The speedup of a system using multiple processors is limited by the fraction of the task that cannot be parallelized. $\text{Speedup} = 1 / ((1 - f) + f/n)$.
CPI	The average number of clock cycles per instruction, used to measure processor performance.
Cache	A small, fast memory that stores frequently accessed data to reduce average memory access time.
Direct-Mapped Cache	A cache organization where each memory block maps to exactly one cache line.
Set-Associative Cache	A cache where each memory block maps to a set of multiple cache lines.
AMAT	The average time to access memory, computed as $\text{Hit Time} + \text{Miss Rate} \times \text{Miss Penalty}$.
Virtual Memory	A memory management technique that gives each process the illusion of a large, contiguous address space.
TLB	A cache for page table entries that speeds up virtual-to-physical address translation.
Superscalar	A processor that can issue multiple instructions per clock cycle.
VLIW	An architecture where multiple operations are encoded in a single long instruction.
RAID	Redundant Array of Independent Disks — a storage technology that combines multiple disk drives for performance and/or redundancy.

Formulas (verbatim) — full sweep

Formula	Statement
CPU Time	$\text{CPU Time} = \text{Instruction Count} \times \text{CPI} \times \text{Cycle Time}$
Speedup	$\text{Speedup} = \text{Old Time} / \text{New Time}$
Amdahl's Law	$\text{Speedup} = 1 / ((1 - f) + f/n)$
AMAT	$\text{AMAT} = \text{Hit Time} + \text{Miss Rate} \times \text{Miss Penalty}$
Offset bits	$\text{Offset bits} = \log_2(\text{Block Size})$
Index bits	$\text{Index bits} = \log_2(\text{Number of Sets})$
Tag bits	$\text{Tag bits} = \text{Address bits} - \text{Index} - \text{Offset}$
Cache Size	$\text{Cache Size} = \text{Number of Sets} \times \text{Associativity} \times \text{Block Size}$
EAT (with TLB)	$\text{EAT} = H \times (t + m) + (1 - H) \times (t + k \cdot n + m)$
EAT (simplified)	$\text{EAT} = H \times m + (1 - H) \times 2m$
TLB Reach	$\text{TLB Reach} = \text{TLB Entries} \times \text{Page Size}$
Page Table Size	$\text{Page Table Size} = \text{Number of Pages} \times \text{Page Table Entry Size}$
RAID 5 Usable	$\text{RAID 5 Usable} = (N - 1) \times \text{Disk Size}$
RAID 5 Write Penalty	$\text{RAID 5 Write Penalty} = 4 \text{ I/Os per write (2 reads + 2 writes)}$

Diagrams · Numericals · Tricky · Top-100

Type	Items
Diagrams	Diagram-Book.md #1–#15 full sweep (canonical)
Numericals	Numerical-Book.md #1/#2/#6/#9 (CPI, CPU time, AMAT, EAT)
Tricky	Top-10-Tricky-Concepts.md all 10
Top-100	Top-100-Questions.md #1–#100

Books · Chapters · Media

Resource	Where
Weeks 1–13 packs	timed past papers

Fear-Killer Pack CA-W14 (verbatim)

Resources: Weeks 1-13 packs • timed past papers

1. List and explain the three fundamental ILP limitations: power wall, memory wall, ILP wall. For each, state one hardware technique that pushes past it and one fundamental physical constraint that remains.
2. Timed multi-cycle datapath draw in 10 min from blank paper. Then 3 pipeline hazard sequence traces. Then 10 hazard timing diagrams — all closed book.
3. Numerical speed drills: retrieve CPI, cache, and virtual memory formulas from memory first, then solve 5 problems per topic without peeking.
4. Sit a full-length final past paper (3 hrs, closed book). Every answer starts from a blank page — no re-reading notes.
5. Redraw any diagram that took >10 min. Repeat until the speed target is met.

Narrative — Week 14 (verbatim)

Topics: Cumulative review of Weeks 1-13

Practice: - Timed full-length past paper (3 hrs, closed book) — START with the past paper, NOT with review - Focus on weak areas identified in midterm — but only via retrieval, NEVER re-reading - Pipeline hazard timing diagrams: 10 problems — all closed book, from blank page - Numerical speed drills: CPI, cache, virtual memory — retrieve formulas from memory first - Multi-cycle datapath: draw in 10 min — timed, closed book, blank paper - FSM design: one complete control unit — from memory - **Spaced repetition:** Redraw any diagram that took >10 min. Repeat until speed target is met.

Retrieval protocol: No passive reading at all this week. Every minute is active retrieval. If you cannot retrieve it, that tells you exactly what to prioritize.

Sleep banking: Continue 9 hrs sleep (bedtime 22:00). Do NOT trade sleep for last-minute studying — it degrades recall by 15-20% per lost hour.

Deliverable: One full past paper solved and self-graded

Time budget: 8 hrs

P0 floor · Drill target · Deliverable · Trap

- **P0 floor:** fear-killer pack pass (Q4 full past paper + Q2 timed draw) · def/formula skim (all W1–W13 defs; formulas from memory first) · same-problem drill #1/#2/#6/#9
- **Same-problem drill target:** every diagram that took >10 min re-drawn until speed target; numerical formulas retrieved from memory before solving
- **Deliverable:** one full past paper solved and self-graded
- **Trap:** no passive reading — every minute active retrieval; START with the past paper, NOT with review

W15 — FINAL EXAM WEEK · 23–29 Nov · Exam

Banner: Rotation CA (Mon+Tue) · Tier FINAL · No new deep study, no floor accrual. Ledger frozen during W15.

Sources

Source	Where	Ref
Narrative	Week-by-Week-Narrative.md → Week 15	lines 273–283
Pack	Fear-Killer-Packs.md → Week 15	lines 118–119
Week manifest	(W15 note at <code>weeks/CA-W14.md</code> L47)	reference

Fear-Killer Pack — Week 15 (verbatim)

Week 15: FINAL EXAM — No new pack. Execution only: numericals → diagrams → theory, label every forwarding path, flag and move on if stuck.

Narrative — Week 15 (verbatim)

Focus: Execution. Do not learn anything new.

- Review 1-page cheat sheet (final version prepared Week 14)
- 8 hours sleep. The hardest numerical paper requires peak cognitive function
- Answer order: numericals → diagrams → theory
- Pipeline hazard tracing: write clearly, label every forwarding path

- If stuck on a problem, flag it and move on. Return if time permits

Exam-day stack (final)

- [] Numerical speed drills first — CPI, cache, VM formulas from memory
- [] Diagram paper: multi-cycle draw (10 min), pipeline hazard traces, every forwarding path labeled
- [] Theory last; flag stuck problems and move on

Notes

- Review the 1-page cheat sheet (prepared Wk14). Sleep 8 h each night.
- Exam order: numericals → diagrams → theory.
- **Ledger frozen during Wk15** — exam window 30 Nov – 18 Dec is a separate phase (per `weeks/CA-W14.md` L47).